DESIGN AND IMPLEMENTATION OF A WEB-BASED ACADEMIC RECORD MANAGEMENT SYSTEM USING PYTHON AND JAVA

Submitted By-

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CERTIFICATE

"Design & Implementation of a Web-Based Academic Record Management System Using Python & Java"

submitted by Tushar Som, a student of BCA 3rd Year, Swami Atmanand Govt. English Medium Model College, during the academic year 2025-26, is a record of the task carried out by me under guidance.

This project is submitted in partial fulfillment of the requirements of Bachelor of Computer Applications (BCA).

The work presented in this project is original and has not been submitted elsewhere for any other degree or diploma.

ACKNOWLEDGEMENT

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ABSTRACT

The '**Academic Record Management System'** also known as **"ARS"** is a web-based application designed to manage student academic records efficiently and securely. The system provides a centralized platform where administrators can manage users and courses, faculty members can enter and update student marks, and students can view their academic performance.

The project is developed using **Python (Flask framework)** for backend processing, **HTML** and **CSS** for frontend design, and **SQLite** for database management. The system implements role-based access control to ensure that different users (Admin, Faculty, and Student) can only access the features appointed to them.

The primary objective of this project is to replace manual record-keeping processes with a structured digital system that improves accuracy, reduces errors, and enhances data accessibility. The system is user-friendly, secure, and can be further extended with additional features such as data analytics and report generation in the future

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CHAPTER 1

INTRODUCTION TO THE PROJECT

1.1 Overview

In the modern education system, maintaining academic records manually has become inefficient and time-consuming. Educational institutions deal with large amounts of student data, course information, and examination records. Manual handling of such records increases the risk of data loss, duplication, and human error.

The Academic Record Management System is a web-based application developed to digitalize and automate the management of academic records. It provides a centralized platform where administrators, faculty members, and students can interact according to their roles and permissions.

1.2 Purpose of the Project

The primary purpose of this project is to design and implement a secure and user-friendly system that simplifies academic record management. The system allows:

- Administrators to manage users and courses
- Faculty members to enter and update student marks
- Students to view their academic performance

The project aims to reduce manual workload and improve efficiency, accuracy, and accessibility of academic data.

1.3 Objectives of the Project

The major objectives of the system are:

- To develop a role-based login system
- To manage student and faculty accounts
- To maintain course and subject records
- To allow secure entry and storage of marks
- To provide students access to their academic records
- To ensure data security and proper database management

1.4 Organization of the Report

This report is organized into different chapters describing the scope, existing system, proposed system, system architecture, implementation, testing, limitations, and future enhancements of the project.

CHAPTER 2

SCOPE OF THE PROJECT

2.1 Scope of the System

The Academic Record Management System is designed to provide a structured and centralized platform for managing academic data within an educational institution. The scope of this project includes the development of a web-based system that supports multiple user roles such as Administrator, Faculty, and Student.

The system allows the Administrator to create and manage user accounts and courses. Faculty members can enter and update student marks, while students can securely view their academic performance.

2.2 Functional Scope

The functional scope of the project includes:

- User authentication through role-based login
- Creation and management of student, faculty, and admin accounts
- Course and subject management
- Entry and updating of student marks
- Viewing academic records based on user role
- Secure storage and retrieval of data using a database

2.3 Limitations of Scope

The current version of the system is limited to:

- Local database usage (SQLite)
- Basic security mechanisms
- No integration with external systems
- No advanced analytics or reporting tools

However, the system can be extended in the future to include advanced features such as online result publishing, graphical analysis of performance, and cloud-based deployment.

CHAPTER 3

EXISTING SYSTEM AND NEED FOR PROPOSED SYSTEM

3.1 Existing System

In many educational institutions, academic records are maintained manually using registers, spreadsheets, or basic computer files. This traditional method of managing records is time-consuming and inefficient. Data entry and retrieval require manual effort, which increases the chances of human error.

Manual systems often lead to problems such as data duplication, loss of records, lack of proper security, and difficulty in generating reports. Additionally, students do not have direct access to their academic records and must rely on administrative staff for information.

3.2 Problems in the Existing System

The major problems in the existing system are:

- Manual handling of records
- High chances of errors in data entry
- Difficulty in searching and updating records
- Lack of centralized data management
- Limited data security
- Time-consuming report generation

3.3 Need for Proposed System

To overcome the limitations of the existing system, a web-based Academic Record Management System is required. The proposed system provides a centralized and digital solution for managing academic data efficiently.

The new system ensures secure role-based access, faster data retrieval, improved accuracy, and better organization of records. It reduces paperwork, minimizes human errors, and enhances transparency between administration, faculty, and students.

The implementation of this system improves overall efficiency and ensures proper maintenance of academic information.

CHAPTER 4

PROPOSED SYSTEM

4.1 Overview of the Proposed System

The proposed system is a web-based Academic Record Management System designed to automate and streamline the management of academic records within an institution. The system provides a centralized platform where administrators, faculty members, and students can perform their respective tasks efficiently based on role-based access control.

The system is developed using Python (Flask framework) for backend processing, HTML and CSS for frontend design, and SQLite for database management.

4.2 Working of the Proposed System

The system operates through a secure login mechanism where users are authenticated based on their assigned roles. After successful login:

- The Administrator can manage users and courses.
- Faculty members can add and update student marks.
- Students can view their academic records.

All data is stored in a structured database and retrieved dynamically when required.

4.3 Advantages of the Proposed System

The proposed system offers several advantages:

- Centralized data management
- Reduced manual workload
- Improved accuracy and efficiency
- Role-based secure access
- Faster data retrieval
- User-friendly interface

4.4 Feasibility of the Proposed System

The system is technically feasible as it uses widely available technologies such as Python, Flask, HTML, CSS, and SQLite. It is economically feasible since it does not require expensive hardware or software. The system is also operationally feasible because it is easy to use and requires minimal training.

CHAPTER 5

SYSTEM ARCHITECTURE

5.1 Overview

The Academic Record Management System follows a client-server architecture. The system is web-based, where users interact with the application through a web browser. The backend processes user requests, communicates with the database, and returns the appropriate response to the user interface.

The architecture ensures proper separation between the presentation layer, application logic, and database layer.

5.2 Architecture Components

The system consists of the following major components:

1. Presentation Layer

This layer is responsible for the user interface. It is developed using HTML and CSS. Users interact with the system through web pages such as Login Page, Admin Dashboard, Faculty Dashboard, and Student Dashboard.

2. Application Layer

This layer handles the core logic of the system. It is developed using Python with the Flask framework. It processes user requests, validates input, manages sessions, and controls role-based access.

3. Database Layer

This layer stores all system data such as user details, course information, and marks. SQLite is used as the database management system. It ensures proper storage, retrieval, and updating of records.

5.3 Data Flow

When a user logs into the system:

1. The login credentials are sent to the application server.
2. The backend validates the credentials using the database.
3. If authenticated, the user is redirected to their respective dashboard.
4. Any data entry or retrieval request is processed through the application layer and updated in the database accordingly.

5.4 Security Mechanism

The system implements role-based authentication to ensure that users can only access features assigned to their role. Session management is used to maintain user login state during system usage.

CHAPTER 6

MODULES DESCRIPTION

The Academic Record Management System is divided into different modules based on user roles and system functionality. Each module performs specific tasks to ensure proper operation of the system.

6.1 Login Module

The Login Module provides secure authentication for users. It verifies user credentials and identifies their assigned role (Administrator, Faculty, or Student). Based on the role, the user is redirected to the appropriate dashboard. This module ensures secure access control within the system.

6.2 Admin Module

The Admin Module allows the administrator to manage the overall system. The main functionalities include:

- Creating and managing user accounts
- Adding and managing course details
- Viewing academic records
- Monitoring system activities

The administrator has full control over user management and course management.

6.3 Faculty Module

The Faculty Module enables faculty members to manage academic records related to students. The key functionalities include:

- Entering student marks
- Updating existing marks
- Viewing academic records

This module ensures that faculty members can securely handle student performance data.

6.4 Student Module

The Student Module allows students to view their academic records. Students can log in to the system and access:

- Their marks
- Course details
- Performance information

This module ensures transparency and easy access to academic information for students.

6.5 Marks Management Module

The Marks Management Module is responsible for storing, updating, and retrieving student marks. It interacts directly with the database to ensure accurate record keeping. The module supports secure data entry and role-based viewing access.

CHAPTER 7

TECHNOLOGY USED

7.1 Introduction

The Academic Record Management System has been developed using modern web development technologies. The selected technologies ensure reliability, efficiency, and ease of implementation. The system follows a web-based architecture using frontend, backend, and database components.

7.2 Frontend Technologies

HTML (Hypertext Markup Language)

HTML is used to design the structure of web pages such as the Login Page, Dashboards, and Record Management pages. It provides the basic layout and structure of the application interface.

CSS (Cascading Style Sheets)

CSS is used to design and enhance the visual appearance of the system. It controls layout, colors, fonts, responsiveness, and overall user interface styling.

7.3 Backend Technology

Python

Python is used as the core programming language for backend development. It handles business logic, authentication, database communication, and routing.

Flask Framework

Flask is a lightweight web framework in Python. It is used to manage server-side operations such as handling user requests, session management, and rendering dynamic web pages.

7.4 Database

SQLite

SQLite is used as the database management system for storing user data, course details, and student marks. It is lightweight, easy to integrate, and suitable for small to medium-scale applications.

7.5 Supporting Language

Java

Java is included as a supporting programming language as part of the project requirements. It demonstrates understanding of object-oriented programming concepts and can be integrated for additional functionality if required.

CHAPTER 8

TESTING AND IMPLEMENTATION

8.1 Implementation

The Academic Record Management System was implemented using a structured development approach. The frontend was designed using HTML and CSS to create a user-friendly interface. The backend logic was developed using Python with the Flask framework, which handles routing, authentication, and database operations.

SQLite was used as the database to store user details, course information, and student marks. The system was developed and tested in a local development environment.

The implementation process included:

- Designing database tables
- Creating login and role-based access functionality
- Developing dashboards for Admin, Faculty, and Student
- Implementing modules for user management, course management, and marks entry
- Testing data storage and retrieval

8.2 Testing

Testing was conducted to ensure that all modules function correctly and meet the project requirements.

1. Unit Testing

Individual modules such as Login, Add User, Add Course, and Add Marks were tested separately to verify the correct functionality.

2. Integration Testing

The interaction between frontend, backend, and database was tested to ensure smooth data flow and correct redirection based on user roles.

3. System Testing

The complete system was tested to ensure that all components work together properly without errors.

8.3 Test Cases

Some of the test cases performed include:

- Valid login credentials → Successful login
- Invalid login credentials → Error message displayed
- Admin adding a new user → User created successfully
- Faculty entering marks → Marks stored correctly in database
- Student viewing marks → Correct records displayed

The system performed successfully for all major test cases.

CHAPTER 9

LIMITATIONS

Although the Academic Record Management System performs its intended functions effectively, there are certain limitations in the current version of the project.

9.1 Technical Limitations

- The system uses SQLite as a local database, which is suitable for small-scale applications but may not support very large datasets efficiently.
- The system is designed for local deployment and does not include cloud hosting or online deployment features.
- Advanced security mechanisms such as password encryption and multi-factor authentication are not implemented in this version.

9.2 Functional Limitations

- The system does not include advanced data analytics or graphical performance reports.
- There is no automated email notification system for results or updates.
- The application does not support multi-institution management.

9.3 Operational Limitations

- The system requires basic technical knowledge for maintenance.
- It is currently intended for use within a single institution environment.

Despite these limitations, the system successfully fulfills its primary objective of managing academic records efficiently and securely.

CHAPTER 10

FUTURE ENHANCEMENTS

Although the Academic Record Management System successfully performs its core functions, several improvements and additional features can be implemented in the future to enhance its performance and usability.

10.1 Security Enhancements

- Implementation of password encryption using advanced hashing techniques.
- Integration of multi-factor authentication for enhanced security.
- Role-based permission customization for more granular access control.

10.2 Functional Enhancements

- Generation of downloadable reports in PDF or Excel format.
- Integration of graphical analysis and performance charts for students.
- Implementation of search and filter options for quick data retrieval.
- Online results publishing system.

10.3 Technical Enhancements

- Migration from SQLite to a scalable database system such as MySQL or PostgreSQL.
- Deployment of the system on a cloud server for online access.
- Implementation of REST APIs for integration with other systems.

10.4 Usability Improvements

- Mobile-responsive design enhancements.
- Improved user interface with modern design frameworks.
- Notification system for important updates.

These enhancements would make the system more scalable, secure, and suitable for large-scale institutional use.

CHAPTER 11

CONCLUSION

The Academic Record Management System has been successfully designed and implemented as a web-based application to manage academic records efficiently. The system provides a centralized and structured platform for handling student data, course information, and marks management.

Through the implementation of role-based access control, the system ensures that administrators, faculty members, and students can access only the functionalities assigned to them. This improves security and maintains proper data organization.

The project demonstrates the practical application of web development technologies such as Python (Flask), HTML, CSS, and SQLite. It successfully replaces manual record-keeping methods with a digital solution, reducing errors and improving efficiency.

Overall, the system meets its objectives and provides a reliable foundation for further enhancements and scalability in the future.

BIBLIOGRAPHY

The following resources were referred to during the development of this project:

1. Flask
   https://flask.palletsprojects.com/

2. Python
   https://www.python.org/

3. SQLite
   https://sqlite.org/docs.html

4. Online HTML & CSS Tutorials

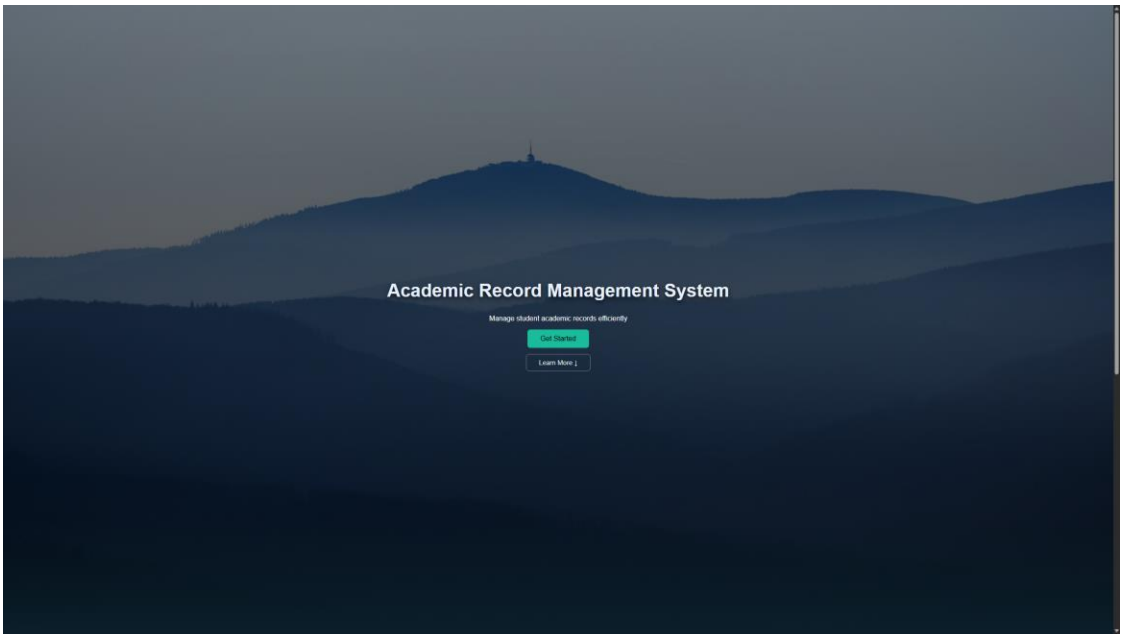
5. And Last Not Least My Programming Notes & Classroom Materials Helped Me Alot...

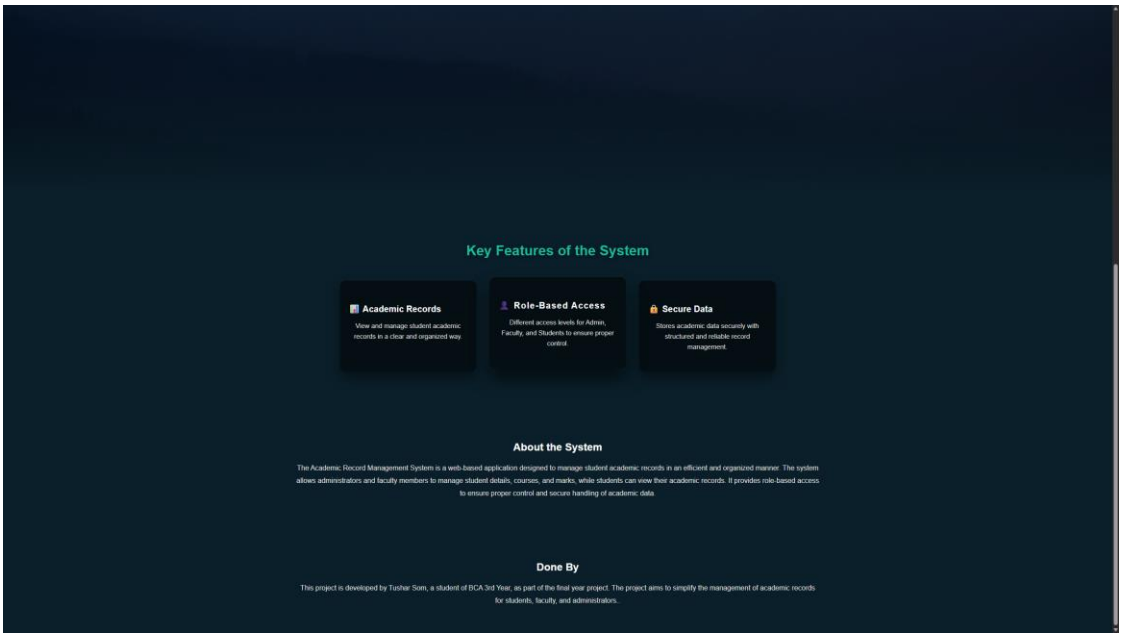
ANNEXURE

SYSTEM SCREENSHOTS

In this section, include all major screenshots of the system with short descriptions below each image

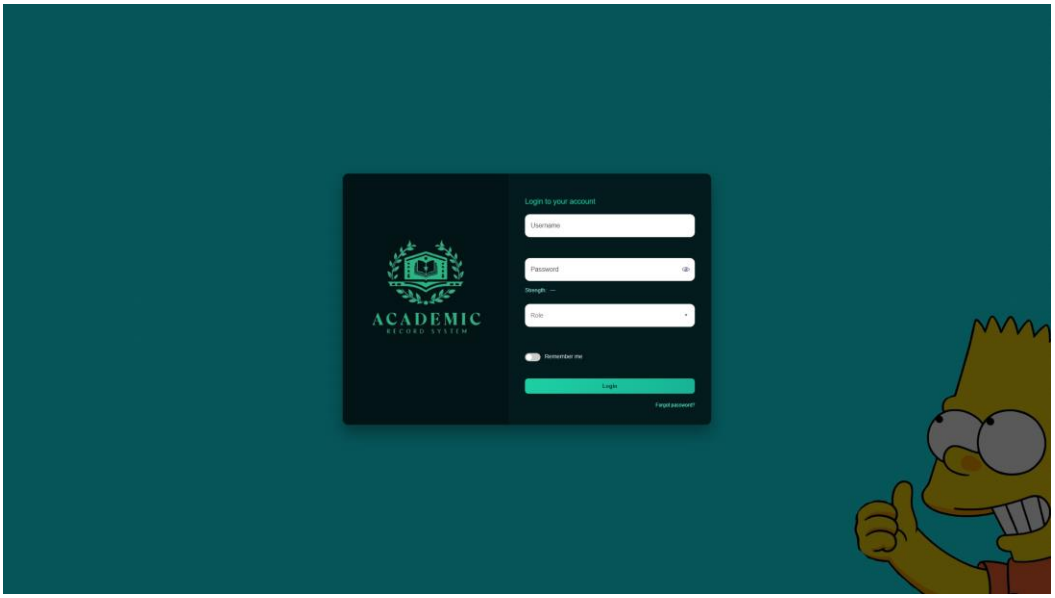
Homepage-

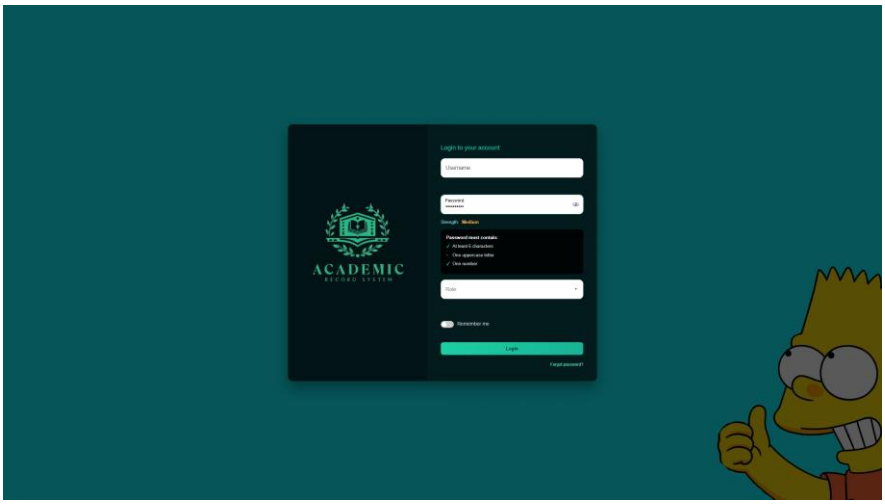


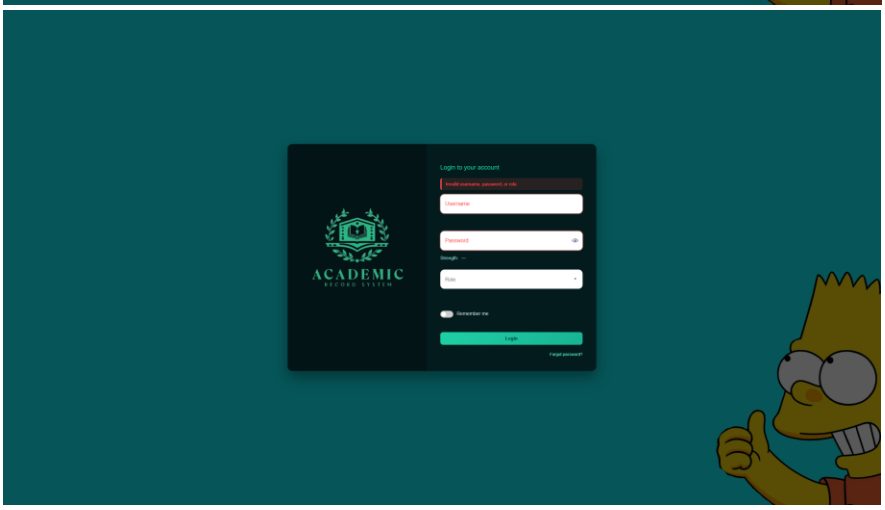


Loginpage-

The Login Page provides role-based authentication for Admin, Faculty, and Student users. It includes username, password, and role selection fields with validation and security features

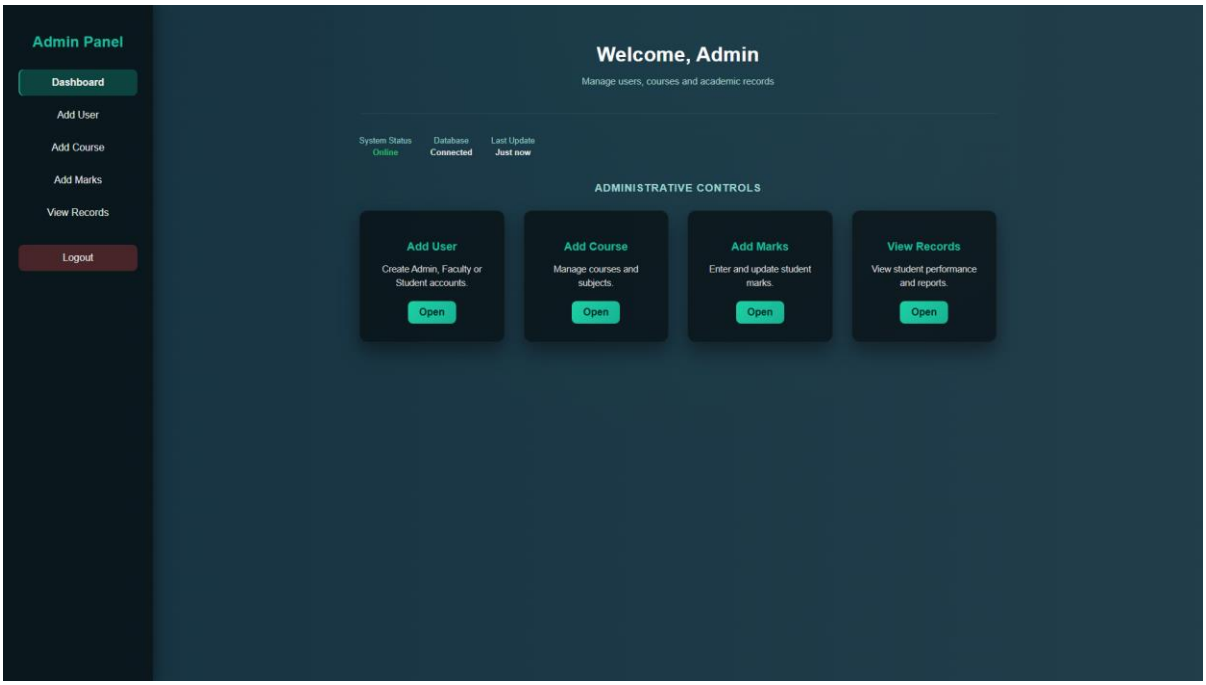






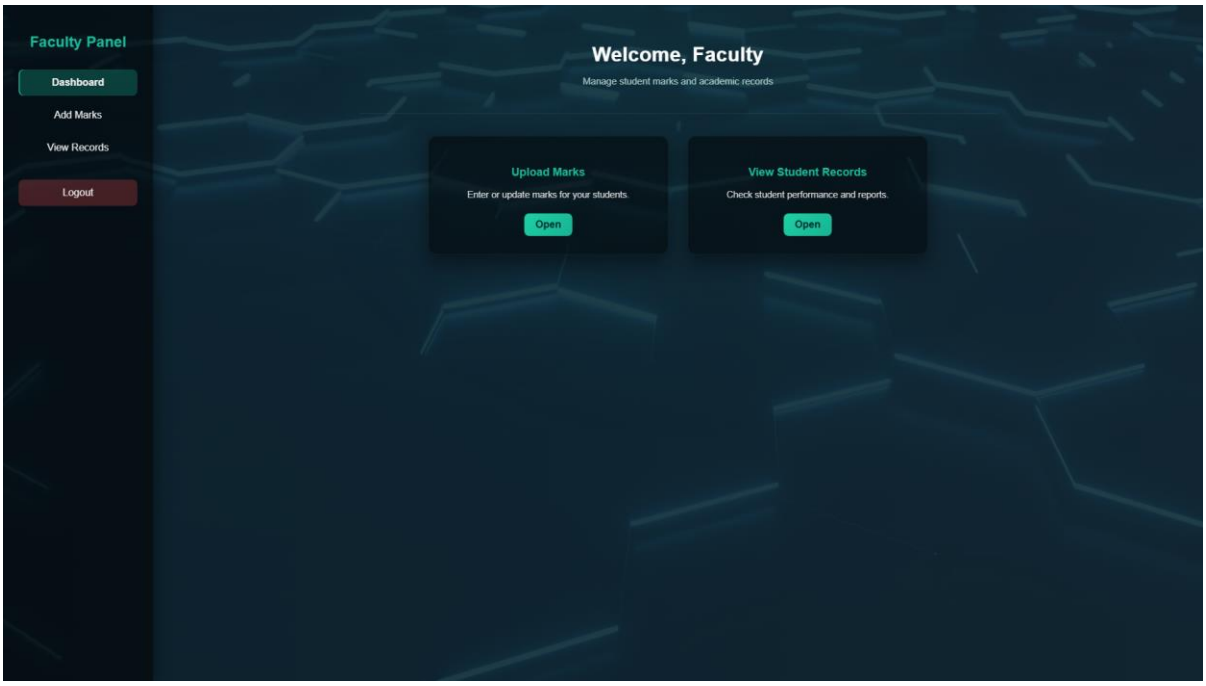
Adminpage-

The Admin Dashboard allows management of users and courses. It provides access to system control features and record viewing options.



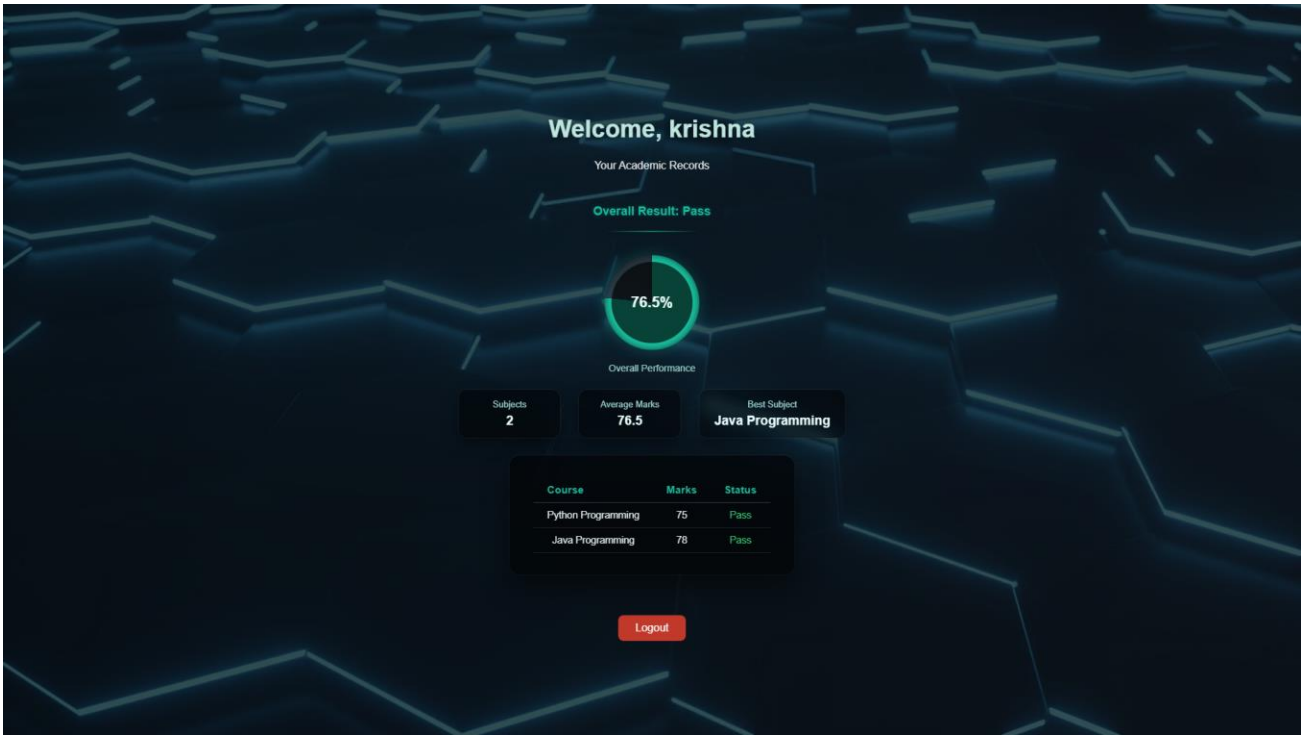
Facultypage-

The Faculty Dashboard enables faculty members to add and update student marks and view academic records.



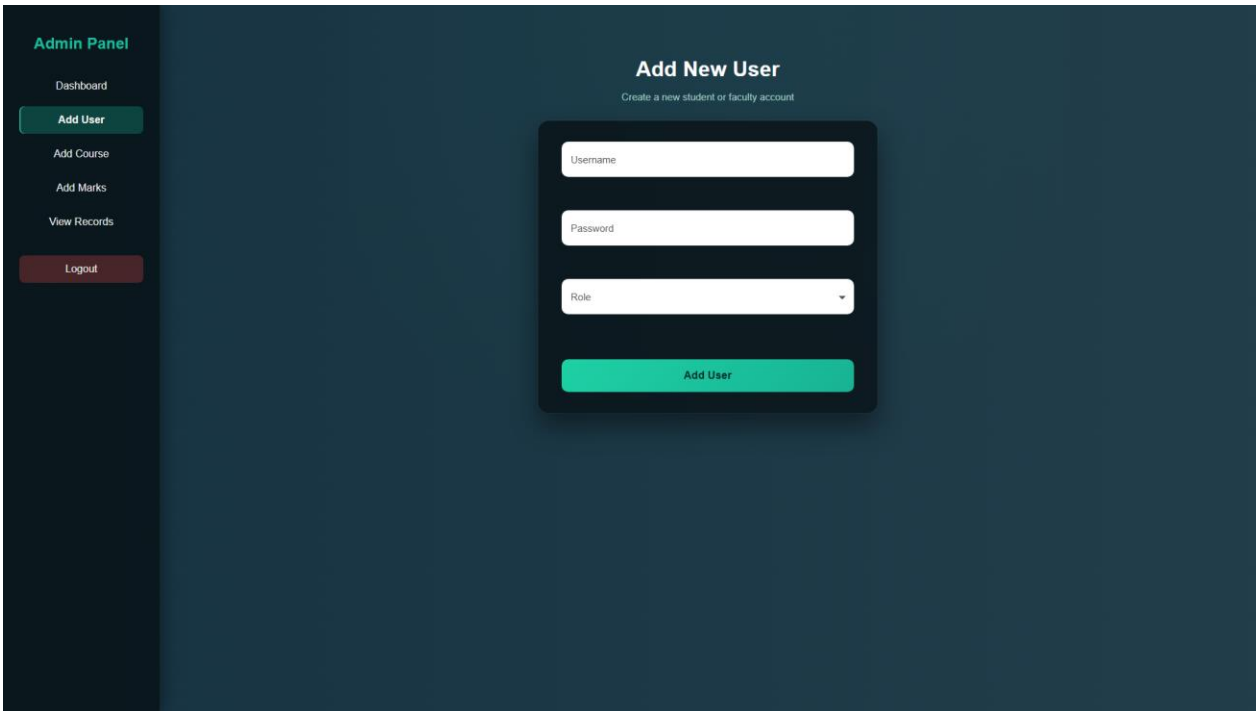
Studentpage-

The Student Dashboard allows students to securely view their academic performance and marks.



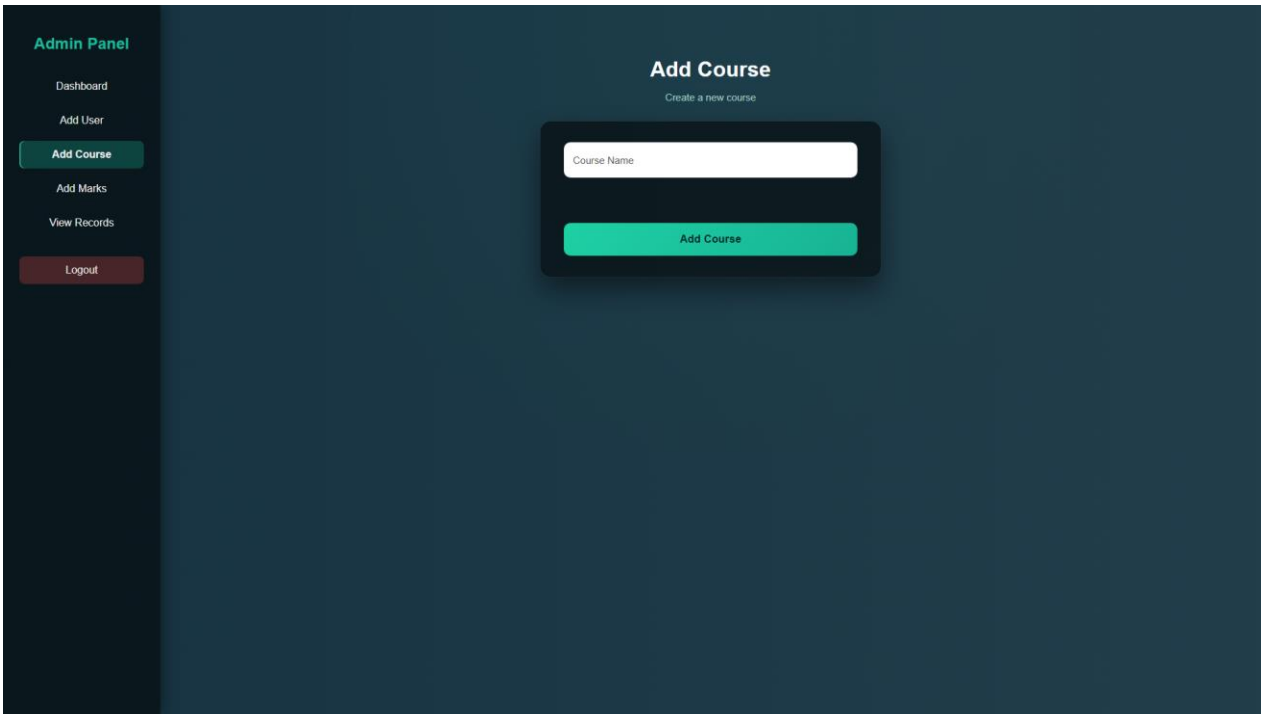
Add_User_Page-

This page allows the administrator to create and manage user accounts within the system.



Add_Course_Page-

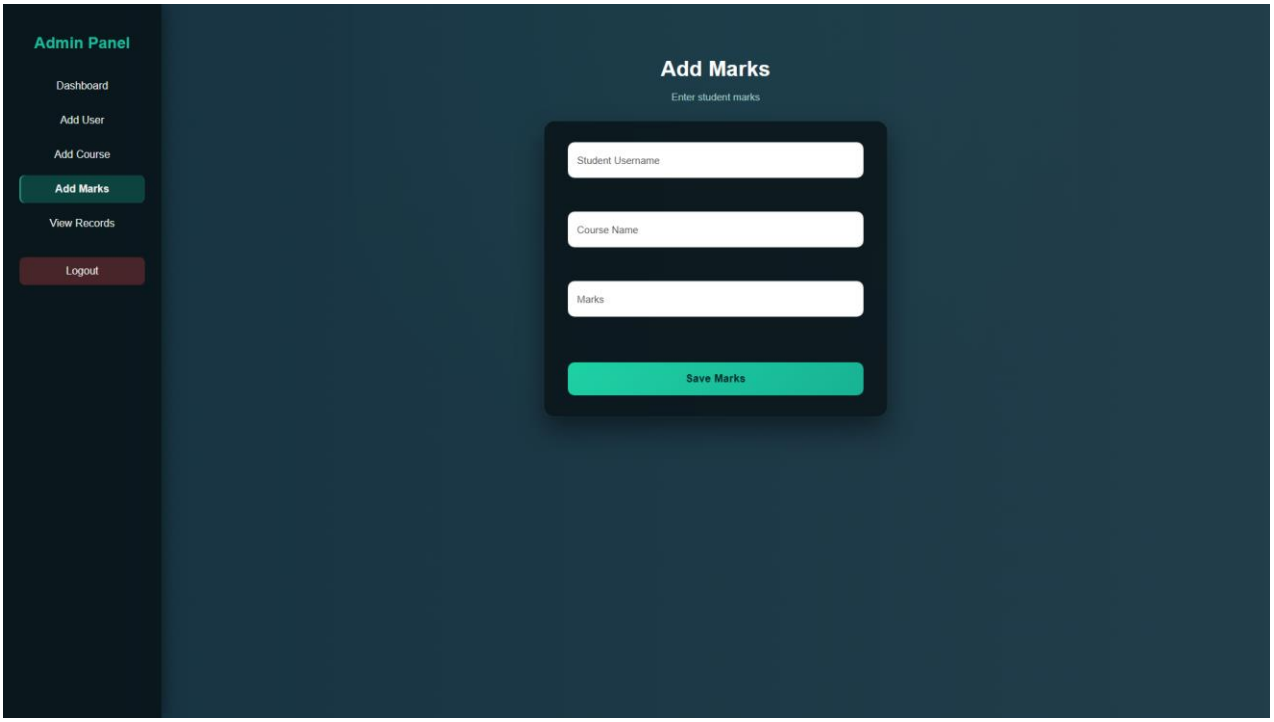
This page allows the administrator to add and manage course details in the system.



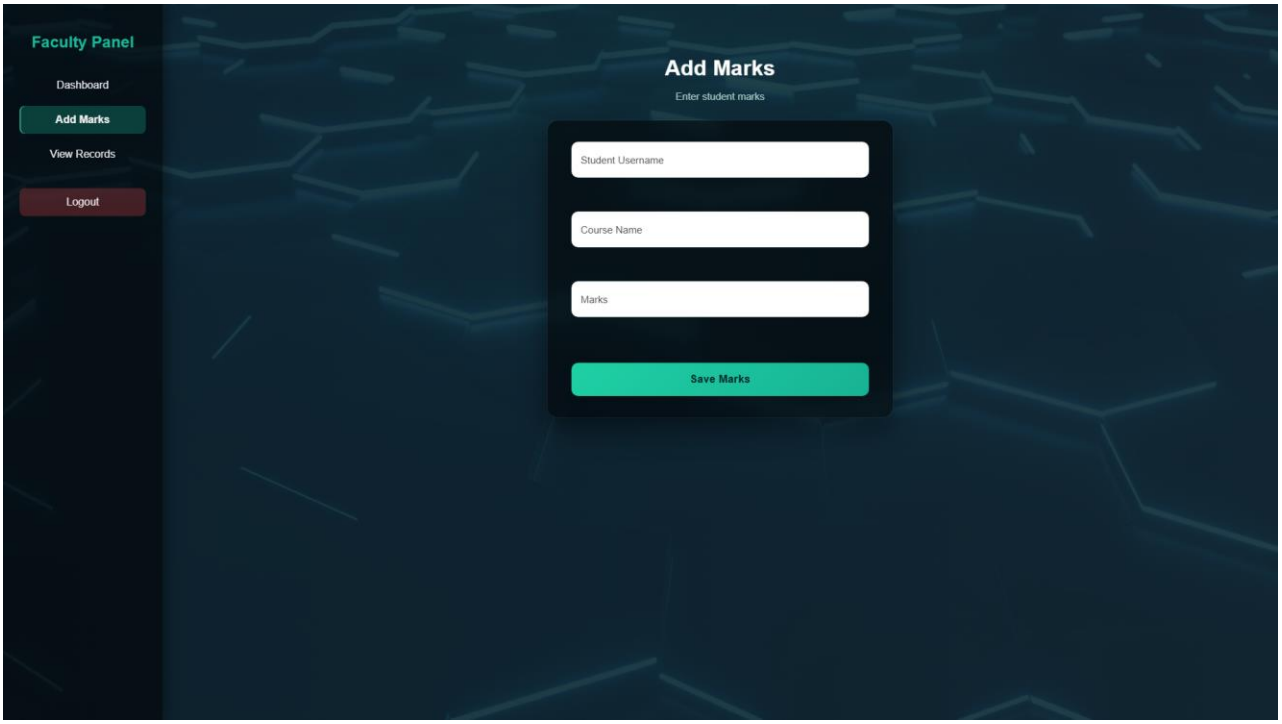
Add_Marks_Page-

The Faculty can enter or update student marks through this interface.

Admin POV-



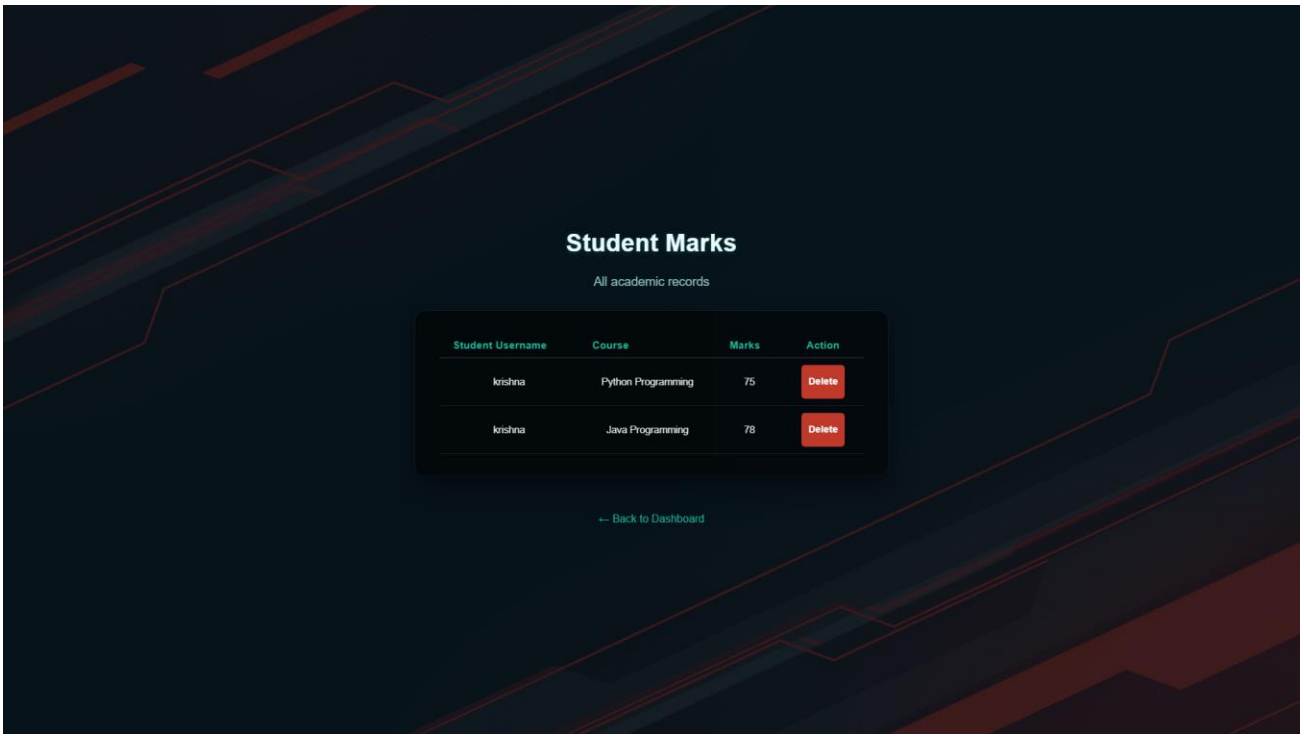
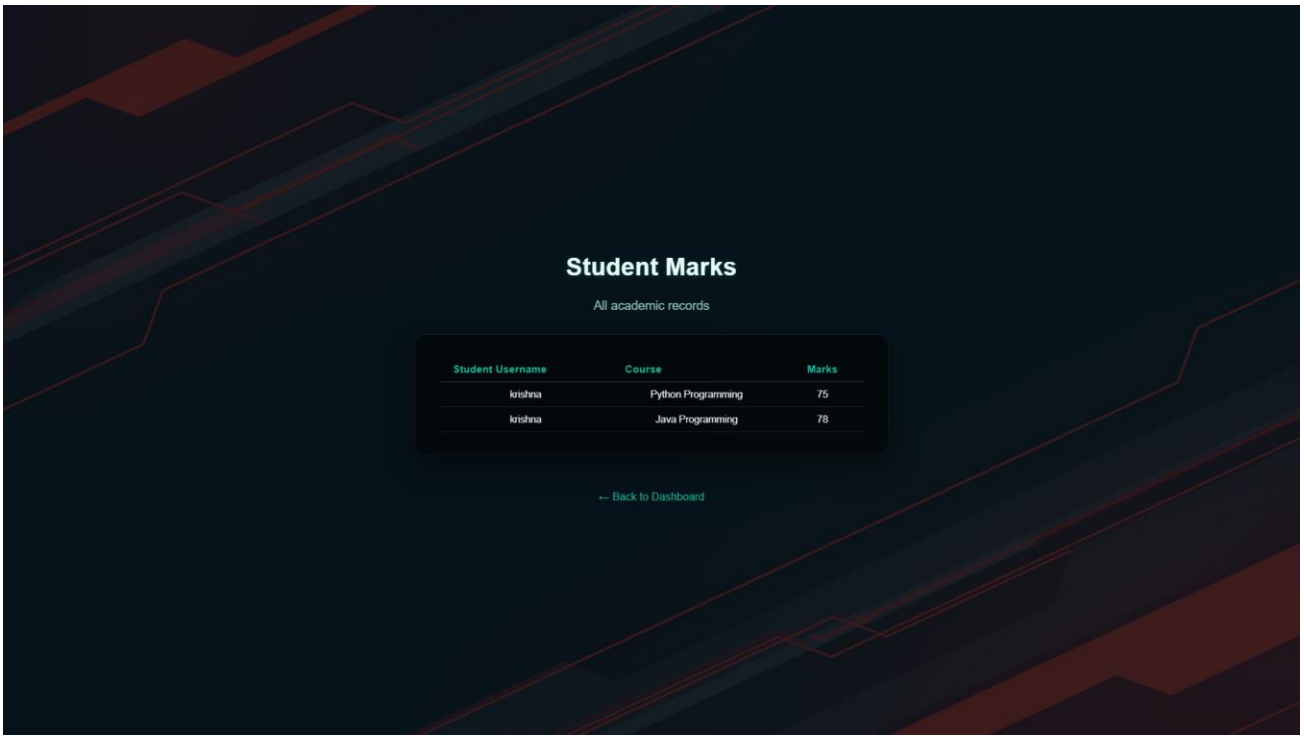
Faculty POV-



View_Records_Page-

This page displays academic records based on the logged-in user's role.

Admin POV- 'Delete Available!!'



Faculty POV-